

Physics LATAM - Riemannian and Complex Geometry — Homework 3 Solutions

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Course: Physics LATAM - Riemannian and Complex Geometry
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Exercise 3.1 — Equivalence of two definitions of vector bundle

Problem. Show the two standard definitions of a rank- r real vector bundle are equivalent: (A) a total space with local trivializations $\phi_i : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$; and (B) an open cover with smooth transition maps $g_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL(r, \mathbb{R})$ satisfying the cocycle conditions.

Solution (construction in both directions).

Direction (A) $\rightarrow$ (B).

1. Let (U, ϕ_U) and (V, ϕ_V) be two trivializations. On the overlap $U \cap V$ both give vector-space isomorphisms of each fibre with $\mathbb{R}^r$. Define the transition map

$$g_{UV}(x) := (p_2 \circ \phi_V) \circ (p_2 \circ \phi_U)^{-1} \in GL(r, \mathbb{R}), \quad x \in U \cap V,$$

where $p_2 : U \times \mathbb{R}^r \rightarrow \mathbb{R}^r$ is projection to the second factor. Concretely, if $\phi_U(v) = (x, v_U)$ and $\phi_V(v) = (x, v_V)$ for $v \in E_x$, then $v_V = g_{UV}(x)v_U$.

2. The functions g_{UV} are smooth because they are compositions of the smooth trivialization maps and projections. They satisfy the cocycle conditions: $g_{UU} = \text{id}$ and on triple overlaps $g_{UV}g_{VW}g_{WU} = \text{id}$ by straightforward composition of the fibre identifications.

Direction (B) $\rightarrow$ (A).

1. Start from the disjoint union $\bigsqcup_{\alpha} (U_{\alpha} \times \mathbb{R}^r)$ with the product smooth structure on each piece and equip it with the quotient topology for the equivalence relation

$$(x, v, \alpha) \sim (x, g_{\alpha\beta}(x)v, \beta) \quad \text{for } x \in U_{\alpha} \cap U_{\beta}.$$

Denote the quotient map by $q : \bigsqcup_{\alpha} (U_{\alpha} \times \mathbb{R}^r) \rightarrow E$ and the quotient space by $E = q(\bigsqcup_{\alpha} U_{\alpha} \times \mathbb{R}^r)$. Define $\pi : E \rightarrow M$ by $\pi([x, v, \alpha]) = x$ (well-defined because relation preserves the base point).

2. To give E a smooth manifold structure we construct charts whose compatibility follows from smoothness of the $g_{\alpha\beta}$. For each index α define

$$\Psi_{\alpha} : U_{\alpha} \times \mathbb{R}^r \longrightarrow \pi^{-1}(U_{\alpha}) \subset E, \quad \Psi_{\alpha}(x, v) = [x, v, \alpha].$$

The map Ψ_{α} is continuous and surjective onto $\pi^{-1}(U_{\alpha})$, and it has inverse given by the assignment $[x, w, \beta] \mapsto (x, g_{\beta\alpha}(x)w)$ on $\pi^{-1}(U_{\alpha})$ (this inverse is well-defined because of the cocycle relations). The inverse is smooth because it is obtained by composing the smooth map $[x, w, \beta] \mapsto (x, w)$ in the local representation with the smooth transition $g_{\beta\alpha}(x)$. Thus Ψ_{α} is a homeomorphism (indeed a diffeomorphism) onto its image and gives the desired local chart.

3. Chart transition maps: on overlaps $\pi^{-1}(U_{\alpha} \cap U_{\beta})$ the coordinate change between Ψ_{α}^{-1} and Ψ_{β}^{-1} is exactly

$$(x, v) \mapsto (x, g_{\alpha\beta}(x)v),$$

which is smooth because $g_{\alpha\beta}$ is smooth. Therefore the collection $\{\Psi_{\alpha}\}$ defines a smooth atlas on E .

4. Smoothness of π : in these charts we have

$$(\Psi_{\alpha}^{-1} \circ \pi \circ \Psi_{\alpha})(x, v) = (x),$$

i.e. π corresponds locally to the projection $p_1 : U_{\alpha} \times \mathbb{R}^r \rightarrow U_{\alpha}$, which is smooth. Hence $\pi : E \rightarrow M$ is a smooth submersion and each Ψ_{α} is a smooth trivialization. The vector-space structure on each fibre is inherited from the second factor and varies smoothly in the charts.

5. This completes the construction: E is a smooth manifold, π is smooth, local trivializations Ψ_{α} are diffeomorphisms, and transition functions are the given $g_{\alpha\beta}$. Thus (B) yields a vector bundle in the sense of (A).

Exercise 3.2 — Line bundle triviality from a nowhere-vanishing section

Problem. Show that a real line bundle $L \rightarrow M$ which admits a smooth section s with $s(x) \neq 0$ for all x is trivial.

Solution.

1. Define a map $\Phi : M \times \mathbb{R} \rightarrow L$ by

$$\Phi(x, t) = t s(x).$$

This map is smooth and fibre-preserving: $\Phi(x, t) \in L_x$ for all (x, t) . It is linear in the second factor and so a bundle map.

2. Show Φ is a vector-bundle isomorphism. Surjectivity: for any $v \in L_x$ the fibre is one-dimensional, so there exists a scalar t with $v = t s(x)$; hence $v = \Phi(x, t)$. Injectivity: if $\Phi(x, t) = 0$ then $t s(x) = 0$ and since $s(x) \neq 0$ we get $t = 0$.
3. The inverse map is explicitly $\Phi^{-1}(v) = (\pi(v), \lambda(v))$, where $\lambda(v)$ is the unique real number with $v = \lambda(v) s(\pi(v))$. The dependence of $\lambda(v)$ on v is smooth because locally, in a trivialization, the section s is represented by a smooth function $f(x)$ that never vanishes on the chart; then in these coordinates one has

$$\lambda(v) = \frac{v_{\text{coord}}}{f(x)}.$$

Since f is smooth and nonzero on the chart, its reciprocal $1/f$ is smooth, so the quotient above is smooth as a function of the local coordinates of v . Therefore Φ^{-1} is smooth, Φ is a diffeomorphism and linear on each fibre. Hence $L \cong M \times \mathbb{R}$ is trivial.

Conclusion. A nowhere-vanishing global section provides a global frame for a line bundle, so the bundle is trivial.

Exercise 3.3 — Poincaré disk isometric to hyperbolic upper half-plane

Problem. Show the unit disk model (Poincaré disk) with metric

$$ds^2 = \frac{4|dz|^2}{(1 - |z|^2)^2}, \quad |z| < 1,$$

is isometric to the upper half-plane model $\mathbb{H} = \{w \in \mathbb{C} : \text{Im } w > 0\}$ with metric

$$ds^2 = \frac{|dw|^2}{(\text{Im } w)^2}.$$

Solution (explicit Möbius map and pullback check).

1. Consider the Möbius transformation

$$w = \phi(z) = i \frac{1+z}{1-z}, \quad |z| < 1.$$

(I write the fraction explicitly to avoid any ambiguity: $\phi(z) = i(1+z)/(1-z)$.)

This map sends the unit disk to the upper half-plane; in particular one checks directly that

$$\operatorname{Im} \phi(z) = \frac{1 - |z|^2}{|1 - z|^2} > 0 \quad (|z| < 1),$$

so indeed $\phi(D) \subset \mathbb{H}$. Two elementary algebraic identities are key:

(i) the derivative

$$\phi'(z) = \frac{2i}{(1-z)^2}, \quad |\phi'(z)| = \frac{2}{|1-z|^2};$$

(ii) the imaginary part of w :

$$\operatorname{Im} \phi(z) = \frac{1 - |z|^2}{|1 - z|^2}.$$

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2. Pull back the upper-half-plane metric via ϕ . One gets

$$\begin{aligned}\phi^* \left(\frac{|dw|^2}{(\operatorname{Im} w)^2} \right) &= \frac{|\phi'(z)|^2 |dz|^2}{(\operatorname{Im} \phi(z))^2} \\ &= \frac{(2/|1-z|^2)^2 |dz|^2}{((1-|z|^2)/|1-z|^2)^2} \\ &= \frac{4|dz|^2}{(1-|z|^2)^2},\end{aligned}$$

which is exactly the Poincaré disk metric. Therefore ϕ is an isometry between the two models.

Remark. This computation shows how conformal (holomorphic) maps can produce isometries between constant-curvature metrics when the conformal factor matches appropriately; the algebraic cancellation above is the essence of the equivalence between the Poincaré disk and half-plane models.

Exercise 3.4 — Christoffel symbols for the round sphere metric

Problem. Let $S^n \subset \mathbb{R}^{n+1}$ be the unit sphere with the metric induced from the Euclidean metric on $\mathbb{R}^{n+1}$. Compute the Christoffel symbols of the Levi-Civita connection in stereographic coordinates (the standard local coordinates on $S^n \setminus \{N\}$).

Solution (use conformal metric from stereographic projection).

1. Stereographic projection from the North pole gives coordinates $u = (u^1, \dots, u^n) \in \mathbb{R}^n$ on the sphere minus the pole. The pullback metric is conformally flat with conformal factor

$$g_{ij}(u) = \lambda(u)^2 \delta_{ij}, \quad \lambda(u) = \frac{2}{1 + |u|^2},$$

where $|u|^2 = \sum_k (u^k)^2$. (This is standard; derive by the projection formula or recall the textbook expression.)

2. (Assume $n \geq 2$ so the coordinate expression below is nontrivial.)

For a conformal metric $g_{ij} = \lambda^2 \delta_{ij}$ we compute the Christoffel symbols directly from the Levi-Civita formula. First compute the partial derivatives of the metric components:

$$\partial_i g_{j\ell} = \partial_i (\lambda^2 \delta_{j\ell}) = 2\lambda (\partial_i \lambda) \delta_{j\ell}.$$

The inverse metric is $g^{k\ell} = \lambda^{-2}\delta^{k\ell}$, so substituting into the general formula

$$\Gamma_{ij}^k = \frac{1}{2}g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$$

gives

$$\begin{aligned}\Gamma_{ij}^k &= \frac{1}{2}\lambda^{-2}\delta^{k\ell}(2\lambda(\partial_i \lambda)\delta_{j\ell} + 2\lambda(\partial_j \lambda)\delta_{i\ell} - 2\lambda(\partial_\ell \lambda)\delta_{ij}) \\ &= \delta^{k\ell}((\partial_i \ln \lambda)\delta_{j\ell} + (\partial_j \ln \lambda)\delta_{i\ell} - (\partial_\ell \ln \lambda)\delta_{ij}) \\ &= \delta_{jk}\partial_i(\ln \lambda) + \delta_{ik}\partial_j(\ln \lambda) - \delta_{ij}\partial_k(\ln \lambda).\end{aligned}$$

This recovers the conformal formula quoted earlier; the simplification used $\partial_i \ln \lambda = (\partial_i \lambda)/\lambda$ and the contraction $\delta^{k\ell}\delta_{j\ell} = \delta_j^k = \delta_{jk}$.

3. Compute the derivatives of $\ln \lambda$ for the stereographic factor. We have

$$\ln \lambda = \ln 2 - \ln(1 + |u|^2), \quad \partial_i \ln \lambda = -\frac{2u^i}{1 + |u|^2}.$$

4. Substitute into the conformal formula to obtain an explicit expression:

$$\boxed{\Gamma_{ij}^k(u) = -\frac{2}{1 + |u|^2}(\delta_{jk}u^i + \delta_{ik}u^j - \delta_{ij}u^k)}.$$

This is the Christoffel symbol in these coordinates. It is symmetric in i, j as required, and one may check it agrees with the geometric fact that geodesics are great circles when transformed back to the sphere.

Remark :

- When $u=0$ (the point antipodal to the pole in coordinates) the Christoffel symbols vanish, which matches the fact that stereographic coordinates are normal at that point (first derivatives of the metric vanish there up to conformal factor symmetry).

Exercise 3.5 — Christoffel symbols for the hyperbolic upper-half metric

Problem. For $H^n(\mathbb{R}) = \{(x^1, \dots, x^{n-1}, y) : y > 0\}$ with metric

$$g = \frac{R^2}{y^2} \sum_{i=1}^{n-1} dx^i \otimes dx^i + \frac{R^2}{y^2} dy \otimes dy = \frac{R^2}{y^2} \delta_{ab} dx^a dx^b,$$

compute the Christoffel symbols.

Solution (conformal-flat calculation).

1. This metric is conformal to the Euclidean metric with conformal factor $\lambda = R/y$. So we may use the same conformal formula as in the previous exercise:

$$\Gamma_{ij}^k = \delta_{jk} \partial_i (\ln \lambda) + \delta_{ik} \partial_j (\ln \lambda) - \delta_{ij} \partial_k (\ln \lambda).$$

2. Compute derivatives of $\ln \lambda$:

$$\ln \lambda = \ln R - \ln y, \quad \partial_{x^i} \ln \lambda = 0 \quad (i = 1, \dots, n-1), \quad \partial_y \ln \lambda = -\frac{1}{y}.$$

3. Using indices a, b, c ranging over $(x^1, \dots, x^{n-1}, y)$, the only nonzero partial derivatives above involve y . Plugging into the formula gives the nonzero Christoffel components (writing the last coordinate index as n to denote y):

- For k corresponding to an x -direction ($k \leq n-1$):

$$\Gamma_{in}^k = \Gamma_{ni}^k = -\frac{1}{y} \delta_{ik}, \quad (1 \leq i, k \leq n-1).$$

- For $k = n$ (the y -direction):

$$\Gamma_{ij}^n = \frac{1}{y} \delta_{ij}, \quad (1 \leq i, j \leq n-1), \quad \Gamma_{nn}^n = -\frac{1}{y}.$$

4. Collecting the nonzero components succinctly (where indices $i, j, k \leq n-1$ run over the x -coordinates and index n denotes y):

$$\begin{aligned} \Gamma_{in}^k &= \Gamma_{ni}^k = -\frac{1}{y} \delta_{ik}, & (1 \leq i, k \leq n-1), \\ \Gamma_{ij}^n &= \frac{1}{y} \delta_{ij}, & (1 \leq i, j \leq n-1), \\ \Gamma_{nn}^n &= -\frac{1}{y}. \end{aligned}$$

For the $n = 2$ case (the familiar upper-half plane), this reduces to the components given in the worked example above.

Exercise 3.6 — Metric compatibility equivalent to parallel metric

Problem. Let g be a Riemannian metric and ∇ a connection. Prove that ∇ is compatible with g (i.e. preserves inner products under parallel transport) iff $\nabla g = 0$ (the metric is covariantly constant).

Solution.

1. Recall the coordinate formula for the covariant derivative of a $(0, 2)$ -tensor g is

$$(\nabla_X g)(Y, Z) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z).$$

2. If ∇ is compatible with g , parallel transport along any curve preserves the inner product. Differentiating the inner product of two parallel vector fields along the curve gives zero; writing that derivative in the form above yields $(\nabla_X g)(Y, Z) = 0$ for all vector fields X, Y, Z . Hence $\nabla g = 0$.
3. Conversely, if $\nabla g = 0$, let $Y(t), Z(t)$ be parallel vector fields along a curve $\gamma(t)$ (so $\nabla_{\dot{\gamma}} Y = 0$ and similarly for Z). Then

$$\frac{d}{dt}g(Y, Z) = (\nabla_{\dot{\gamma}} g)(Y, Z) + g(\nabla_{\dot{\gamma}} Y, Z) + g(Y, \nabla_{\dot{\gamma}} Z) = 0 + 0 + 0 = 0.$$

Hence parallel transport preserves inner products; this is exactly metric compatibility.

Therefore the two formulations are equivalent.

Exercise 3.7 — Induced connection on tensor product preserves induced metric

Problem. Let (M, g) be Riemannian and consider the induced metric on $T^*M \otimes T^*M$ defined by $g(T, S) := T_{ij}S_{kl}g^{ik}g^{jl}$. If a connection ∇ on T^*M is compatible with g , show the induced connection on $T^*M \otimes T^*M$ is compatible with the induced metric.

Solution.

1. The induced connection on the tensor product is the Leibniz (product) connection: for T, S tensor fields,

$$\nabla_X(T \otimes S) = (\nabla_X T) \otimes S + T \otimes (\nabla_X S).$$

2. Compute the derivative of the inner product:

$$\begin{aligned} X(g(T, S)) &= X(g^{ik}g^{j\ell}T_{ij}S_{k\ell}) \\ &= (\nabla_X g^{ik})g^{j\ell}T_{ij}S_{k\ell} + g^{ik}(\nabla_X g^{j\ell})T_{ij}S_{k\ell} \\ &\quad + g^{ik}g^{j\ell}(\nabla_X T_{ij})S_{k\ell} + g^{ik}g^{j\ell}T_{ij}(\nabla_X S_{k\ell}). \end{aligned}$$

3. Because ∇ is compatible with g on T^*M , we have $\nabla_X g^{ik} = 0$ (equivalently $\nabla_X g_{ik} = 0$ and raising indices commutes with ∇). Therefore the first two terms vanish and we get

$$X(g(T, S)) = g(\nabla_X T, S) + g(T, \nabla_X S),$$

which is exactly the condition that the induced connection is metric-compatible on $T^*M \otimes T^*M$.

This completes the proof.

Exercise 3.8 — Trace and covariant derivative identity

Problem. Let (M, g) be a Riemannian manifold with Levi-Civita connection ∇ . For a $(2, 0)$ -tensor T prove that for any vector field X ,

$$X(\text{tr}_g T) = g^{ij}(\nabla_X T)_{ij},$$

where $\text{tr}_g T = g^{ij}T_{ij}$ is the trace using the inverse metric g^{ij} .

Solution.

1. Compute the derivative of the trace using the product rule:

$$\begin{aligned} X(\text{tr}_g T) &= X(g^{ij}T_{ij}) \\ &= (\nabla_X g^{ij})T_{ij} + g^{ij}X(T_{ij}). \end{aligned}$$

2. Express $X(T_{ij})$ in terms of covariant derivatives:

$$(\nabla_X T)_{ij} = X(T_{ij}) - T_{kj}\Gamma_{i\ell}^k X^\ell - T_{ik}\Gamma_{j\ell}^k X^\ell,$$

where the last two terms are the Christoffel contributions; here we write the connection coefficients as $\Gamma_{i\ell}^k$ and contract them with the components X^ℓ of the vector field X .

Since $\nabla g = 0$ we have $(\nabla_X g)_{ij} = 0$, and equivalently $\nabla_X g^{ij} = 0$ (raising/lowering indices commutes with ∇). Contracting the displayed expression with g^{ij} therefore removes the connection correction terms, leaving exactly

$$X(\mathrm{tr}_g T) = g^{ij}(\nabla_X T)_{ij}.$$

3. This identity is a direct and useful consequence of metric compatibility of the Levi-Civita connection.
